

Fork / join

When a testbench needs parallel activity

When the parent resumes

- Picture two forked threads: A takes thirty time units, B takes ten
- Under join, the parent resumes at thirty , the maximum
- Under join_any, the parent resumes at ten , the minimum
- Under join_none, the parent resumes at zero while both children continue
- That difference drives real testbench structure

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Fork / join sketch

Spawn two threads and see when the parent resumes under `join`, `join_any`, and `join_none`.

Starter example: A takes 30, B takes 10 — step time and watch when the parent resumes under each join flavor.

Load starter example

Challenges (1/22)

Quiz: `join`. Wait for all children? Answer: `join`

Answer

Show hint

Check

Next

Idle

quiz-join

quiz-any

quiz-none

quiz-disable

starter

preset-join

preset-any

preset-none

start

step

join-resume

any-resume

none-resume

b-first

disable

demo

explain

mode-any

code-join

code-any

reset

full-path

Core ideas

join

Wait for *all* — parent @ `max(durations)`.

join_any

Wait for *first* — parent @ `min`; others may linger.

join_none

Do not wait — parent resumes at once.

Browser lab starter

learn_sv_tb — fork join

Real SystemVerilog practice

- In the real SystemVerilog track, open this module's examples prompts
- Restate when you would pick each join flavor , all done, first done, or no wait
- Sketch the fork block below on paper and write the resume time for each keyword with A
- Optional stretch
- ``systemverilog // fork-join , parent waits for ALL children (max time) fork begin #30

Pitfalls to watch

- Do not assume `join_any` stops the other threads
- Do not use `join_none` when the parent must observe child results before continuing
- Do not forget that resume time depends on the slowest or fastest child, not an average
- And remember

Your turn

- Complete the checklist for at least one track , preferably both
- In the browser, load starter, step through join and join_any
- On paper, write a fork block with two delays and label where the parent prints under all
- When you are ready